DH Table - FANUC LR Mate 200iD

Fanuc Course Project Part 1

1. Joint rotation axes

Axis of each joint at the zero pose

Joint	Axis	Direction
J1	0 0 1	Vertical (z)
J2	0 1 0	Horizontal (y)
J3	0 -1 0	Horizontal (-y)
J4	-1 0 0	Along forearm (-x)
J5	0 -1 0	Horizontal (-y)
J6	-1 0 0	Along flange (-x)

2. DH table

From the link dimensions + axes above:

i	θi	di (mm)	ai (mm)	αi (deg)	offset (deg)
1	q1	330	50	-90	0
2	q2	0	330	0	-90
3	q3	0	35	-90	0
4	q4	335	0	90	0
5	q5	0	0	-90	0
6	q6	80	0	0	0

- offset of -90deg on joint 2 so upper arm is vertical when q2 = 0

- J4, J5, J6 axes meet on one point (a4 = a5 = d5 = 0) sphere wrist

3. Joint limits in the DH model

With DH frames, joint 3 to 6 rotate the opposite way compared to the ROS-Industrial model, so:

q_DH = [q1, q2, -q3, -q4, -q5, -q6]

Joint	ROS model (deg)	DH model (deg)
J1	-170 to 170	-170 to 170
J2	-100 to 145	-100 to 145
J3	-70 to 205	-205 to 70
J4	-190 to 190	-190 to 190
J5	-125 to 125	-125 to 125
J6	-360 to 360	-360 to 360

4. Converting from the FANUC teach pendant

J3 on pendant is measured from base, not J2. Using pendant angles:

q_DH = [J1, J2, -(J2 + J3_pendant), -J4, -J5, -J6]

5. Checks (to do in MATLAB)

1. All joints = 0deg, so fkine should give us flange at (465, 0, 695) mm

2. Zero pose -> flange z-axis should point forward (+x)

3. Compare fkine with the 30 test cases from mattj23 using the conversion in section 4. They use FANUC world frame (330 mm higher) so we need to subtract 330 mm from our z.

Sources

ROS-Industrial, lrmate200id_macro.xacro (commit d8f42bd, 2025-02-19). https://github.com/ros-industrial/fanuc/blob/noetic-devel/fanuc_lrmate200id_support/urdf/lrmate200id_macro.xacro

ROS wiki, fanuc_lrmate200id_support package page.
https://whiteoak.umd.edu/roswiki/fanuc_lrmate200id_support.html

FANUC America, LR Mate 200iD product page.
https://www.fanucamerica.com/products/robots/series/lr-mate/lr-mate-200id

mattj23 on GitHub, lrmate-200id (test_cases folder). https://github.com/mattj23/lrmate-200id

Robot-Forum, "How to get the calibrated Fanuc DH parameters from Fanuc Teachpendant".
https://www.robot-forum.com/robotforum/thread/51369-how-to-get-the-calibrated-fanuc-dh-parameters-from-fanuc-teachpendant/

Peter Corke, Robotics Toolbox for MATLAB. https://petercorke.com/toolboxes/robotics-toolbox/